

Project Report on

‘Online Retail Sales Analytics Dashboard using Power BI’

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Chapter 1: Problem Analysis Introduction

1.1 Background of the Study

The current competitive and information-oriented business world is very competitive, with the organizations constantly generating large amounts of sales transaction data both online and in-store. Each and every purchase, every single return and every interaction with a customer adds up to the huge amounts of data that, with appropriate use, can provide meaningful information about the business performance and customer habits. Nonetheless, even though there is a lot of data, raw and unprocessed information in itself does not give any substantial value to the decision makers. Such data will be in a complexity and hardly analytical, thus restricting their application in strategic planning and operational decision making unless it is properly organized, analyzed and visualized.

The growing dependency on data to grow the business has given rise to the Business Intelligence (BI) tools that are essential in converting the raw data into useful and understandable and actable insights. Such tools as Power BI would allow organizations to operate with massive amounts of data effectively and display them in the form of interactive dashboard and visual reports. Such dashboards can make the complex datasets easier to understand, since they are turned into a graphical model such as charts, graphs and key performance indicators (KPIs), which give the decision makers the opportunity to understand key trends and patterns in a short period of time.

The project is aimed at examining an online sale of products data and creating an informative Power BI dashboard to facilitate the proper business decision making. The dashboard will offer an interactive and easy-to-understand display of the important business indicators such as revenue patterns, best-selling products, customer distribution, and operational details. The dashboard provides the user with a way to explore and analyze information effectively by displaying data in a graphical format that is easy to understand.

In this way, the context of this paper indicates the importance of data analytics and business intelligence tools in retail business that is increasing.

1.2 Problem Statement

The number of sales transactions that retail organizations undertake on a daily basis is enormous and,

therefore, analyzing them manually is a complicated and inefficient process. Due to the growth of businesses and customer interactions, the volumes of data are growing exponentially and the organizations are finding it increasingly challenging to manage and interpret information using the conventional means. The analysis of large datasets using manual methods is likely to produce errors, delays and inconsistencies as well as eventually influence the quality in decision making.

Decision makers have to deal with the issue of ineffectiveness of the sales data to be clearly and structured represented. In the case of raw or unstructured data storage, it becomes hard to distinguish the key performance indicators and derive meaningful information. To illustrate, in the absence of visualization, it can be very difficult to identify the best performing products, to comprehend sales differences between the seasons, or to be able to determine how customers buy their products. This confusion may lead to business opportunities and poor business strategies.

The other important problem is that the traditional systems are not able to offer real-time insights. Fixed reports and spreadsheets do not enable the user to dynamically work with data and interact with it. As such, decision makers tend to use obsolete information, which cannot be appropriate in the present business environment. Such time lag in accessing the relevant data may impact decision making and decrease the efficiency of the organization.

Thus, the necessity to develop a data-driven solution to work with large amounts of sales data efficiently, displaying it in an unambiguous format, interactive, and in a manner of visual appeal is high. A business intelligence dashboard is a powerful remedy to this issue as it means that data analysis and visualization are combined in a single platform. This type of a system allows the user to interact with data in real-time, detect patterns and trends and make decisions in a timely manner.

This project will deal with these problems by creating a Power BI dashboard that will convert raw retail sales data into valuable insights. The proposed system will improve the process of decision-making and the overall performance of the business by offering interactive visualizations and instant analysis features.

1.3 Objectives of the Project

The primary aim of this project is to examine the retail sales data and create an interactive business

intelligence dashboard on Power BI that will enable efficient and well-informed decision making. To enable this, the project is oriented towards a number of specific objectives that help in creation of whole picture of business performance.

Among the main goals, there is the analysis of sales data to draw the trends and patterns of revenue on a monthly basis. The analysis of the fluctuations of the sales on a time basis will allow companies to learn about the seasonal changes and organize their strategies in these respects. The analysis aids organizations to predict demand, to optimize inventory and to allocate resources efficiently.

The project will also focus on examining the performance of the customers distribution and geographic sales. Knowing the location of the customers and the sales per region helps the business to adjust their marketing plans and open up new markets. Geographic analysis can offer great value to the organization in terms of regional preferences, and also helps the organization determine areas that are likely to record high growth.

Through this analysis, businesses are able to know the preferences of customers and see which sales channel is effective. Through the evaluation of the various channels, companies are able to streamline their sales mechanisms, as well as enhance customer relation.

Lastly, the project will provide all findings through proper visualization applications including charts, KPIs, filters, and slicers. These tools make the dashboard usable and more interactive, giving individuals the freedom to learn more about the data intuitively and in a highly adaptable way. The project shows viability in the practice of business intelligence tools in the retail sales analysis by accomplishing these goals.

1.4 Scope of the Project

The scope of the given project will focus on the analysis of retail sales with the use of one dataset where the key elements will be to convert raw data into meaningful results with the help of data preprocessing and visual analysis methods. The project will have several phases that include cleaning and preparing data, then analysis and development of dashboard.

Data preparation is a mandatory component of the project and it is done in Python in the Google Colab environment. This stage entails the management of missing values, duplication and data consistency.

The quality and accuracy of the analysis is greatly enhanced by the ability to prepare the dataset ahead of time. The data is then cleaned and structured and then imported into power BI to continue the analysis and visualization.

Some of the components incorporated in the Power BI dashboard created in this project are key performance indicators (KPI), line charts, bar charts, maps, and donut charts. These graphics visual representation is employed to describe various features of the dataset in a straightforward and succinct way. The interactive capabilities like filters and slicers enable the users to tailor their view and examine data in relation to certain specifications.

This project has a number of key areas that are analyzed. The sales trend is analyzed to determine how the revenues fluctuate with time whereas the product performance analysis determines the highest selling items and categories. The customer insights are obtained and used to determine the purchasing behavior and distribution trends. The geographic analysis will provide data on sales operation in various regions and the operational logistics are examined in order to determine the returns and the sales channels.

It should be mentioned that the given project is confined to the analysis of the provided data set, and it is not accompanied by the real-time data integration or predictive analytics. The emphasis is more on descriptive analytics, which is a summarization and interpretation of past data. Irrespective of this shortcoming, the project is a good illustration of how business intelligence tools can be applied to analyze data and assist in decision making.

1.5 Review of Existing Systems

The conventional sales reporting systems have been using manual tools like spreadsheets and static reports in analyzing data. Although these have been popularly applied in the past, they are usually time consuming and not flexible enough to be used in contemporary business set ups. Entry and analysis of data manually elevate the chances of errors and inconsistencies and thus organizations are finding it hard to trust the accuracy of their reports.

The fact that traditional systems do not give real-time insights is one of the key limitations. The static reports are usually created with a certain interval meaning that in most cases decision makers have to use the information that is outdated. This latency when reaching the recent data may be a problem in

the timely decision making and lower the overall efficiency of the business processes. Moreover, the interaction with the data and the possibility to discover the patterns and trends are not offered by the use of only static reports.

The second problem related to the traditional systems is the problem of dealing with large datasets. With the growth of the number of data, it becomes progressively difficult to manage and analyze information through spreadsheets. This tends to cause low processing time and low productivity. Companies can find it hard to derive any meaningful insights on their data, and they are therefore missing out on opportunities and making suboptimal decisions.

Modern business intelligence applications, Power BI, Tableau, and Google Data Studio, on the contrary, also provide sophisticated solutions to data analysis and visualization. These applications offer interactive dashboards that allow users to interactively explore data in real time, filter data and create customized reports. This facilitates comprehending the complex datasets and extracting important insights as the visual components like charts and graphs are used.

The system that will be applied in this project makes use of Power BI to break the constraints of the traditional way of reporting. The system provides a more effective and easier mode of interpreting data as it incorporates both data analysis and visualization within the platform. It improves the clarity and accessibility of the information by using the KPIs and interactive charts enabling the decision makers to identify trends and make informed decisions in a short amount of time.

To sum up, the analysis of the current systems shows that modern BI tools are applied in the contemporary data-driven business environment. The adoption of interactive dashboards instead of the manual reporting practices is a major advancement in the manner in which organizations evaluate and use data. The project shows how Power BI can be successfully applied to the problem of dealing with the threats posed by the traditional systems and better the overall decision-making.

Chapter 2. Project Design and Methodology

2.1 System Overview

The main goal of this project is to examine data on retail sales and provide business information useful

in action with the help of the interactive Power BI dashboard. The system has been planned to convert raw transactional data to a significant and visually friendly analytical tool that will help the business stakeholder to make informed decisions. The general system structure follows a designed data analytics pipeline that consists of four consecutive stages, namely, data collection, data preprocessing, data analysis, and data visualization.

At the first stage, raw data on retail transaction is obtained through online retail sales dataset which is the main input into the system. In its raw state, this data is not consistent, missing some entries, and its attributes are not uniform that makes them inappropriate to analyze directly. As a result, the dataset passes a rigorous preprocessing phase during which the data quality concerns are addressed systematically. The main tool used in this stage of preprocessing is Python, which is being run in the Google Colab software. The preprocessing tasks involve elimination of duplicate records, treatment of missing values, date format standardization and extraction of new analytical attributes like Revenue column -calculated as the product of Quantity and Unit Price, and temporal attributes like Year and Month, derived out of attribute of invoice date.

After the preprocessing is complete, the processed and enriched data is then exported to CSV and, then, loaded into Microsoft Power BI Desktop where the analytical dashboard is to be built. The Power BI platform gives us a powerful analysis application of data visualization tools that allow one to create dynamic and interactive visual reports. The resultant dashboard provides detailed information in a variety of retail performance measures such as sales patterns over time, product-level performance indicators, customer distribution trends, and geographic sales trends, and operational efficiency indicators. This set of components can collectively offer an integrated and end-to-end data analytics solution that effectively seals the divide between the raw transactional data and strategic business intelligence.

The dashboard consists of eight pages: Page 1: Overall business performance overview, Page 2: Product performance & returns analysis, Page 3: Customer and regional revenue insights, Page 4: Logistics and shipping efficiency analysis, Page 5: Profitability and cost analysis, Page 6: Customer segmentation and behavior analysis, Page 7: Revenue trends and growth analysis, Page 8: Forecasting and advanced business insights.

2.2 System Architecture

The proposed retail sales analytics project system architecture is designed in such a manner that it

includes four different functional layers with each having a particular set of responsibilities within the end-to-end data processing and visualization chain. Such a stratified structure guarantees the clear separation of the concerns, the adoption of modularity and the ease of tracing the data flow between the raw input and the final business insights.

The lowest level of the system is the first layer which is called the Data Source Layer. It includes the original online retail sales data, which holds transactional data that summarizes different characteristics of customer purchase. It is this raw data that is used as the single source of truth upon which all the successive computations are extracted.

The second layer is the Data Processing Layer which works with the raw dataset and performs the transformation and cleaning of the raw data into a clean and structured and analytically prepared format. This layer takes advantage of Python programming in the Google Colab unit to conduct a detailed list of preprocessing functions. These functions include data cleaning, removing duplicates, imputing missing values, and type conversion, and feature engineering, which are all the necessary conditions to analyze anything meaningful.

The third level is known as the Data Visualization Layer, which takes the refined and sanitized dataset and implements it in the Microsoft Power BI Desktop and creates the interactive analytics dashboard. This layer is in charge of converting structured data into extensive visual representations such as bar charts, line graphs, KPI cards, maps, and drill-through reports; which together create the analytical interface of the system.

The fourth and last layer is the User Interaction Layer which is the interface where the end users interact with the dashboard. Interactive features like filters, slicers, and dynamic visualizations to enable the user to view the information in various ways and derive context-specific information are offered to users. The entire architecture guarantees a smooth and effective flow of information with the raw data all the way to actionable business intelligence, making the system scalable and useable in practice.

2.3 Proposed Methodology

The method employed by this project is a systematic and step wise data analytics process which entails the process of acquiring raw data to business insights generation. The methodology is logically organized in the sense that the successful selection of the next stage will depend on successful accomplishment of the previous stage therefore making the results of the analysis correct and true.

It begins with Data Collection phase in which one acquires online data of sales in terms of retailing. This data will be the primary source of the transactional data which will contain the records regarding the purchase of a separate product and in this case, the information about the invoice number, the product description, the quantity, the unit price, customer identifiers, and geographic data will be presented.

The second step is Data Cleaning and Preprocessing that is performed in the framework of Google Colab in Python. It is at this stage that data is explored into quality irregularities like null values, duplicates, data type and formatting inconsistencies.

The third step is Feature Engineering; it involves elaboration of the derived attributes, in order to add value to the information value of the data. One of the columns is especially the Revenue which is computed as the product of the Quantity and Unit Price columns of the transactions. Moreover, Year and Month are calculated based on invoice date so that time-series and period-based analysis could be done. Such created attributes add the dimensionality to the data set and enable more granular insights.

The final phase is Insight Generation that entails the utilization of the finalized dashboard in the analysis and interpretation of the trends of sales performance, customer behavior and operational efficiency. Under this stage, the outcome of the methodology is the accumulated knowledge that is applied in making a data-driven decision. The overall methodology is depicted through the following flow:

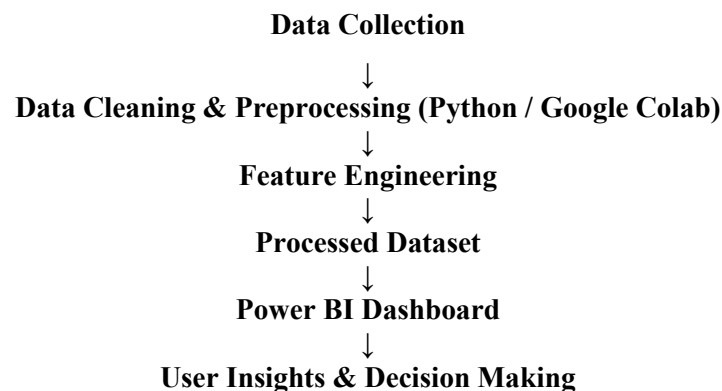


Figure 2.1: Proposed Methodology Flowchart

2.4 Algorithms Used

As opposed to traditional software development projects, which are based largely on machine learning models or sophisticated models of computation, the current project largely uses data aggregation and analysis methods, which are very appropriate to the character of a Business Intelligence (BI) program. It is based on deriving meaningful summaries and comparisons of structured data as opposed to predictive or prescriptive modelling.

Basic statistical operations like summation, count, average, and grouping of data are the fundamental analytical operations to be used in this project. Those operations are utilized in calculating key performance indicators (KPIs) in the Power BI environment including the Total Revenue, Total Orders, Total Customers, and Average Order Value. The calculation of Total Revenue, e.g. can be seen as the summation of all the separate revenue values of individual transactions, and the calculation of Average Order Value can be seen as the division of total revenue by the number of unique orders.

Another important element of the analytical framework is sorting and filtering algorithms. Such algorithms are important to determine the best-selling products and countries, that is, to sort the data by descending revenue or order volume, and thus allow performing a comparison of performance. Filtering operations are also used to isolate certain subsets of data according to user-specified conditions, e.g. time ranges, product line, or geographic area.

Data grouping techniques are used to divide the dataset into various categorical dimensions such as product category, sales channel and geographic region. The system enables performance differences to be identified at segment levels depending on the shared attributes that allow grouping of records in a given segment of the retail business. Although the methods are fairly easy to apply in their respective application, they all form a complete and profound analytical paradigm that is beneficial to the goals of the project.

2.5 Tools and Technologies Used

The integrated application of multiple software tools and technologies, all of which perform specific functional tasks in the project pipeline, made the successful creation of the retail sales analytics dashboard possible.

The initial tool used in the data preprocessing and data cleaning was Python, which was run on the Google Colab cloud-based development environment. Google Colab was chosen because it is easy to

use, it allows teams to work together, and it easily integrates with Python libraries. The application of the Pandas library, an open-source, popular data processing and analysis tool in Python, was specifically used to execute a multivariate variety of data transformation tasks such as filtering, merge, type-conversion and generating derived columns.

The key structure of data visualization and dashboard development was Microsoft Power BI Desktop. The Power BI was selected due to its comprehensive range of visualizations, inherent ability to support interactive filtering and drill-through functionality, and its support of Data Analysis Expressions (DAX) - a formula language, in which one can create custom calculated measures and key performance indicators. DAX was greatly applied during the project and was employed in the definition of the metrics, Total Revenue, Total Orders, Total Customers, as well as the Average Order Value.

The processed data was saved and moved between environments in CSV (CommaSeparatedValues) format, which is a lean, platform-independent format, fit to be imported to Power BI. The Microsoft Bing Maps which are embedded within the Power BI was used to support the geographical visualization of sales data so that regional and country-wide sales distributions could be represented on interactive map pictures.

2.6 Experimental Setup and Environment

The experimental set-up to the current project was geared towards supporting both the preprocessing and the visualization of data, in which both cloud-based and desktop-based software tools were combined.

Preprocessing of the data was carried out within Google Colab which is a cloud platform based on Jupyter Notebook. In this environment there is no local Python environment needed and it has scalable computing utilities that can be accessed through a regular web browser. The preprocessing scripts were written and executed in this environment and then the final clean data exported into a CSV file was generated upon the completion of the scripts.

The Microsoft windows operating system was used in a local development system in the creation of the visualization and dashboard. This system was deployed Power BI Desktop, and it served as the principal application in the development of dashboard and generation of reports. Hardware requirements of the development system were a typical laptop processor and minimum of 8 GB of

memory that was sufficient to support the data loading, transformation and visualization processes that were performed in Power BI.

This stored the data in CSV format throughout the project life cycle which made the preprocessing environment in Google Colab compatible with the visualization one in Power BI. The interactive nature of the power BI dashboards including slicers, filters and drill through reports were tested using the desktop application to ensure responsiveness and accuracy of the analytical reports.

2.7 Dataset Description

The dataset employed in this project is an online retail sales dataset which is a transaction-level record of buying product in an e-commerce platform. The records in the dataset are related to individual product purchase transactions and are linked to an overall set of attributes capturing the commercial, logistical, and customer-specific aspects of every sale.

This data set contains a wide variety of different attributes, such as Invoice Number that is a unique identifier of every transaction; Product Description that defines the essence of the purchased object; Quantity that defines the number of units bought; Unit Price that determines how much one unit would cost. The Customer ID and Country are customer specific features, which can be used to analyze the customer distribution and geographic sales trends. Such operational features as Payment Method, Shipping Cost, and Return Status give the information about the transactional and logistical peculiarities. The dataset also has a Category attribute as well as Sales Channel attribute allowing product-level and channel-level segmentation analyses.

The derived feature calculated on each transaction was $\text{Revenue} = \text{Quantity} \times \text{Unit Price}$ to improve the analytical value of the dataset. This derived column gives a direct value of the monetary worth of each transaction and is the key measurement of sales performance analysis. In addition, univariate elements such as the Year and Month were also derived as part of the Invoice Date attribute to enable the analysis of trends over time and comparisons of periods.

The enriched data that is obtained gives a complete base on the multi-dimensional analysis that is going to be done in this project since one can now explore the trend of sales, the performance of the products, the buying behavior of customers, the geographical distribution of the sales and how effective the different operational dimensions are. The data was treated and authenticated before being inherited in the Power BI dashboard and this guarantees that the entire analytical outputs are based on

precise and trustworthy information.

Chapter 3: Project Implementation

3.1 Development Environment

This project was implemented with the help of a mix of both data processing and visualization tools. The raw data was first processed and cleaned in Google Colab by writing Python code where relevant preprocessing, including the treatment of missing values, data type conversion, and field consistency, was done. The filtered data was then loaded into the Microsoft power BI Desktop which was the main tool to be used to create interactive dashboards.

Data modeling, calculated measures and visualization were done using Power BI. Such functions as DAX (Data Analysis Expressions) were used to calculate such important metrics as total revenue, total orders, average order value, and the total units sold. There were also the standard requirements of the development environment, which comprised of a windows-based operating system, adequate memory needed to run the datasets, and an internet connection to present the maps. This was a combination of tools that guaranteed effective data transformation and quality visual analytics.

3.2 Module Description

The project is organized in eight key modules, which are presented in Power BI in the form of a dashboard page, and each module addresses various analytical aspects of the dataset.

The initial module, Overview Dashboard, gives a summary view of business performance at the high level. It contains several key performance indicators that include total revenue, total orders, average value of orders, total number of customers, and the number of units sold. It also gives the trends of revenue on monthly basis, distribution of revenue by categories, use of maps to show geographical analysis and comparison of sales channel.

Product Analysis Dashboard is the second module that concentrates on product level insights. It shows the top 10 products in terms of revenue, category level unit sales and graphical representation of product returns. The module assists in recognizing high performing product and return behavior.

The third module is Customer and Sales Insights Dashboard which is a detailed insights on customer contributions and sales distribution. It covers the best customers as per revenue, revenue breakdown in the various countries and review of various payment modes. This module aids in comprehending the customer behavior and the regional performance.

The fourth module is Operational Analysis Dashboard, which is concerned with operations efficiency and logistics. It involves the analysis of shipment providers, average cost of shipping by provider and distribution of orders on priority basis. This module offers the knowledge about performance in delivery and cost containment.

The fifth module is Profitability Analysis Dashboard which is concerned with the relationship between cost, revenue and profit. It contains a comparative analysis of the costs versus revenue versus profit in terms of visual representation, contribution of the profit by various categories, and a hierarchical break up of profit by category, country and sales channel. The module aids in determining the most lucrative segments and the role of various factors in the total profitability.

The sixth module is Customer Segmentation Dashboard which focuses on customer groupings in terms of value and behavior. It divides customers into high value, medium value and low value groups and displays their distribution, spending patterns and order behaviour. The module further contains rich customer level data and segmental spending analysis, that assist in comprehending customer contact and targeting measures.

The seventh module is Time-Series and Growth Analysis Dashboard, which concentrates on analyzing the patterns of revenue over time. It encompasses moving average analysis to even out the fluctuations, monthly percentage growth of revenues to monitor the changes in performance and category-wise revenue changes across various time periods. This module offers the understanding of the seasonal patterns and growth consistency and variations of performance over time.

The eighth module is Forecasting and Advanced Analytics Dashboard that offers predictive and strategic information. It involves forecasting of revenues based on past trends, the major players that affect revenue performance, comparison of projected revenues and actual revenues and has a scenario analysis option where variables like change in price can be manipulated. The module aids in making decisions ahead of time and in analyzing possible business results.

3.3 Code Implementation

The implementation of the code was divided mainly into preprocessing the data in Google colab with the help of Python, and analytical modeling with the help of Power BI. During the preprocessing phase, the dataset was loaded in Python libraries, including pandas, where the structure of the dataset and the data cleaning operation were performed. The missing values have been addressed accordingly, unnecessary columns were deleted where appropriate and the data types have been standardized to allow compatibility with Power BI.

The dataset was preprocessed and then exported and loaded into Power BI. In Power BI, the calculated measures were done through the use of DAX which were used to compute the important measures like total revenue (SUM of revenue), total orders (COUNT of invoice numbers), average order value, and total units sold. Additional transformations and aggregations were done directly in Power BI to facilitate different visualizations.

And filters and slicers were introduced to allow interactivity such that users can explore data dynamically according to parameters like country, year and category. Top N filtering, Positioned filtering and advanced filtering techniques were employed to emphasize important entities like top customers and top products. In general, the procedure implemented guaranteed a smooth transition between data preprocessing and visualization.

3.4 System Interface / Screenshots.

The system interface will be an interactive multi-page dashboard in the Power BI, and the dashboard will have four primary pages. The page structure is presented in such a way that it brings clear and easy-to-understand information in the form of charts, maps, and major indicators.

1. The opening page describes the business overview in the form of KPI cards, monthly revenue trend line chart, a revenue bar chart by category, a revenue geographical map by country, and a donut chart of sales channels. On this page, a summary on overall performance is provided.

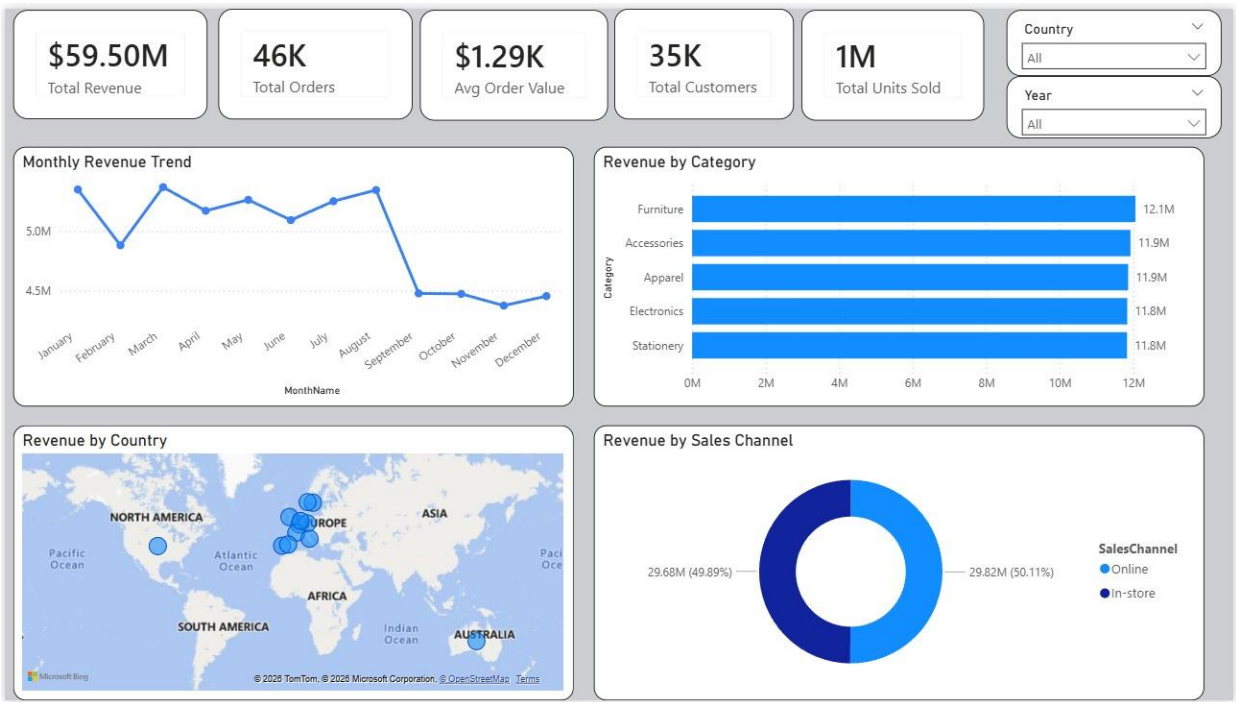


Figure:1

2. The second page is dedicated to the analysis of the product, which consists of the presence of a bar chart of the 10 best-selling items in terms of revenue, a donut chart of the distribution of product returns, and a column chart of the number of units sold according to category. The page assists in determining performance and return trends of products.



Figure:2

3. The third page offers customer and regional information on the form of a bar chart of top customers, a revenue distribution chart by country and a donut chart of payment methods. On this page, preference of customer contribution and payment is emphasized.

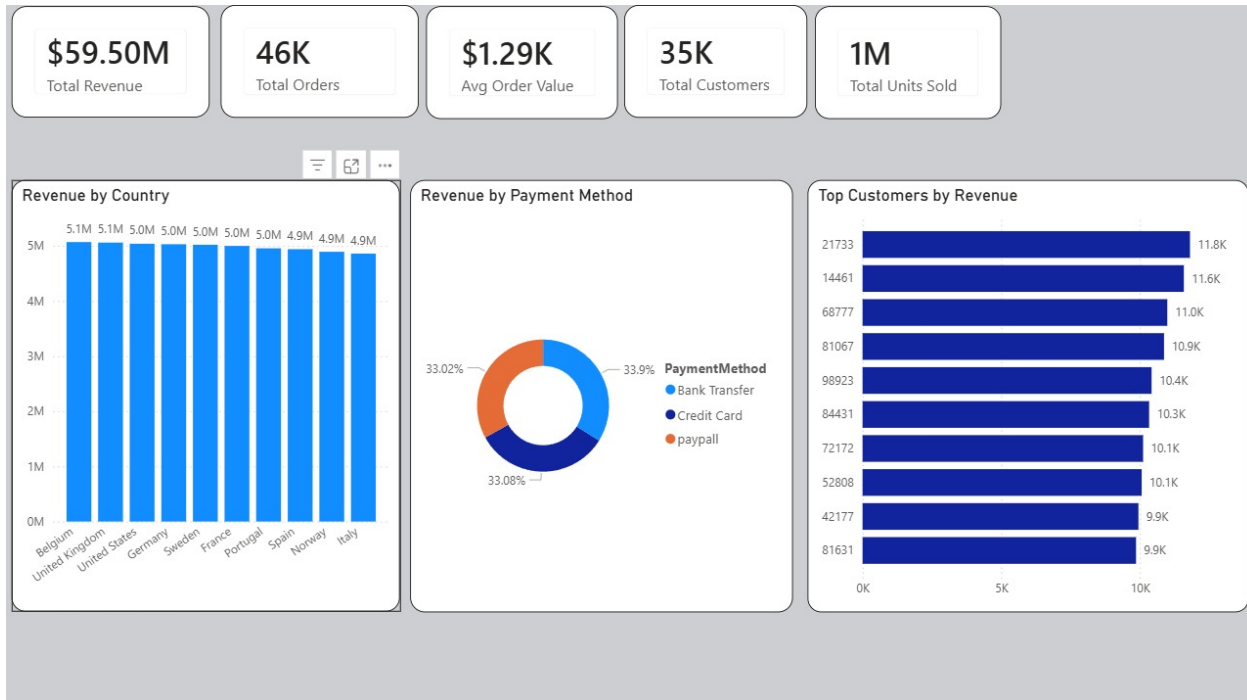


Figure:3

4. The fourth page is dedicated to the working processes and has such points as the performance of shipment providers, the comparison of the average shipping cost, and the distribution of orders by priority. The page assists in the interpretation of the efficiency of logistics and trends of operations.

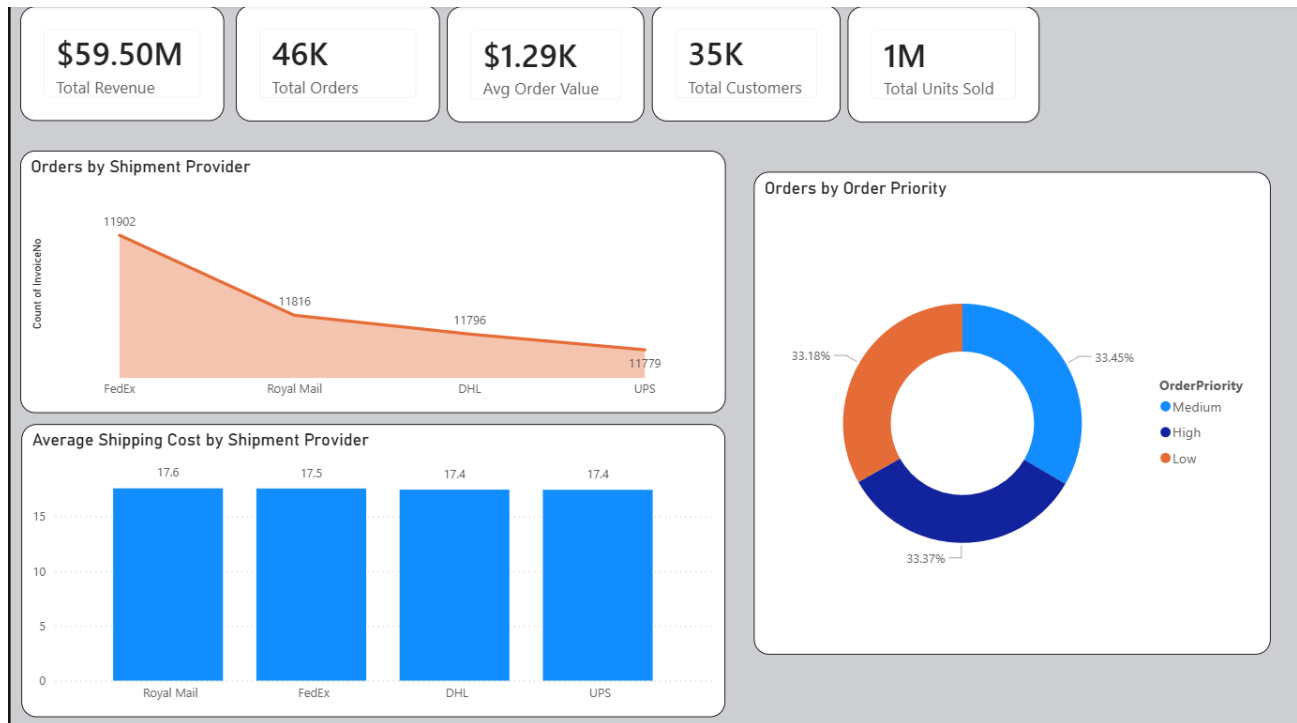


Figure:4

5. The aim of this page is to assess the efficiency and cost effectiveness of the logistics operations. It has KPI cards that consist of the total profit, profit margin and total shipping cost in order to give a quick performance overview. Scatter plot is used to depict the correlation between cost of shipping and revenue per product category and to determine cost-intensive but low-payoff segments. Also, a waterfall chart shows the contribution of each category to profitability.

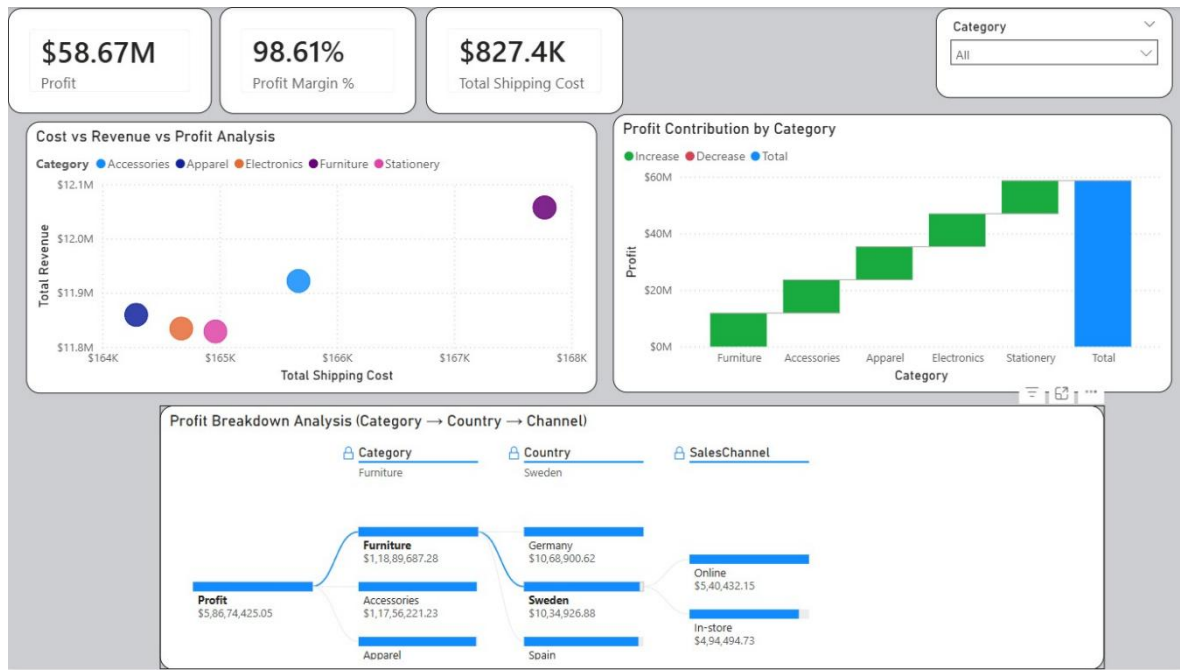


Figure:5

6. The sixth page shows customer segmentation and behavior analysis, which has KPI cards such as total customers, AOV, and high-value customers. It has a donut chart of segments of customers, a scatter plot of customer behavior (orders vs spend), a detailed customer table and a bar chart of total spend by segment. The following page brings out distribution of customer values and buying habits.



Figure:6

7. The seventh page is devoted to the analysis of the trends in revenues and growth, which will include the KPI cards: total revenue, total orders, profit margin, and profit. It has a moving average line chart, a monthly revenue growth chart, and a multi-line chart of category-wise revenue trends. This page assists in establishing trends, growth patterns and seasonality.

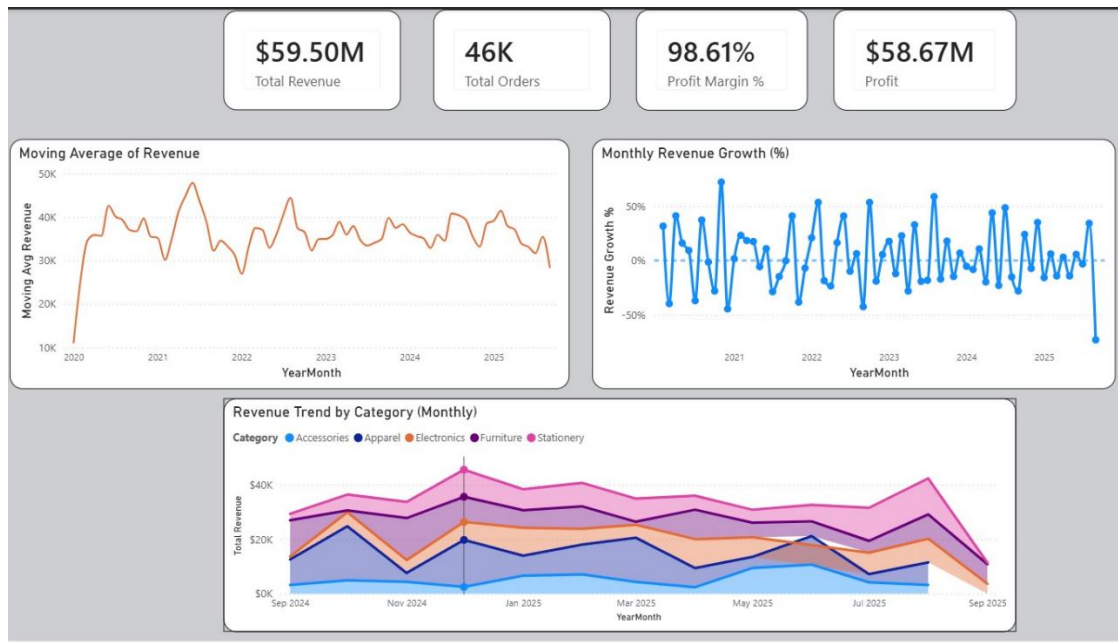


Figure:7

8. The eighth page offers business and forecasting, such as the key influencers and the top segments that impact revenue. It comprises of a forecast chart of future revenue patterns, a gauge visual of projected and actual revenue, and a scenario analysis tool of manipulating variables such as price. This page is helpful in predictive analysis and decision-making based on data.

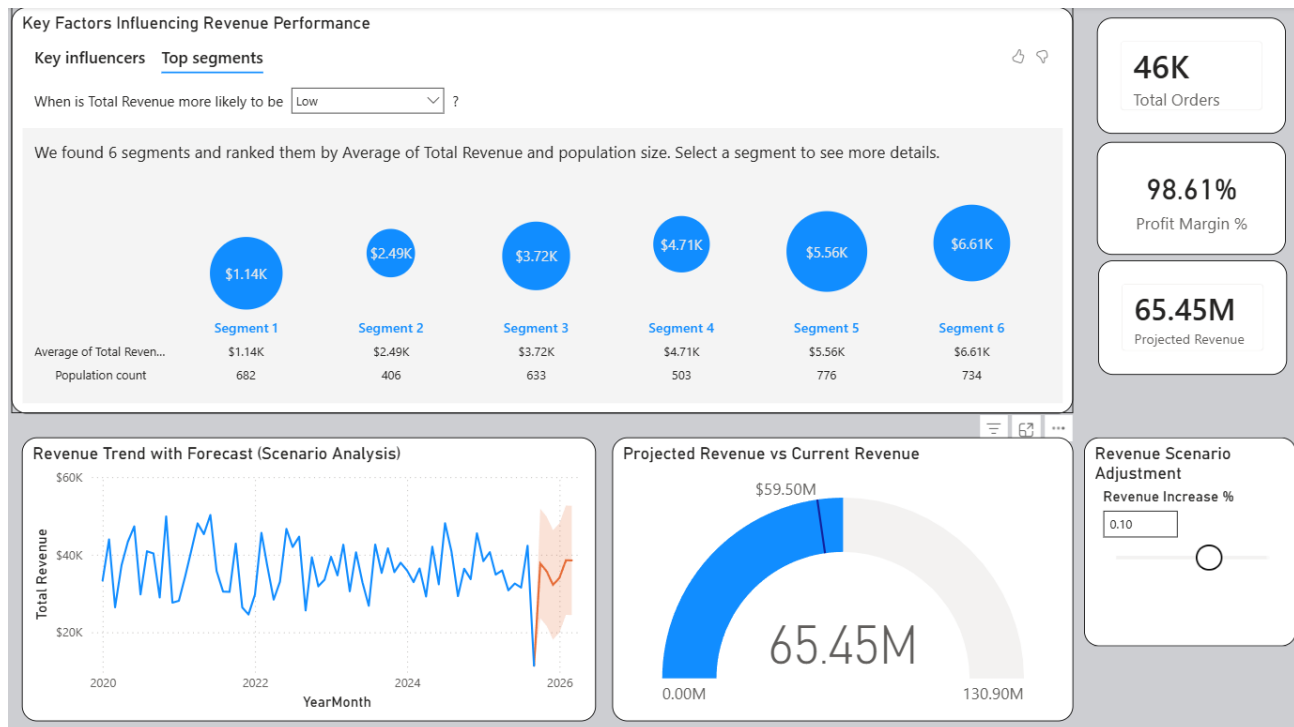


Figure:8

Chapter 4: Project Results and Discussion

4.1 Experimental Setup

The project experimental design entailed integrating information preprocessing and visualization aids to process an online sales data set. The dataset was initially treated with the help of Google Colab where data cleaning and transformation operations were done to maintain accuracy and consistency. This involved the process of dealing with missing data, data type validation, and the formation of the dataset to analyse it.

The data was then processed then imported into Microsoft Power BI to be visualized and analysed. There were several measures that were developed based on the computation of the key measures using DAX, including the total revenue, total orders, average order value, and total units sold. The

dashboards have been developed over eight pages with each page having a certain analytical approach, such as overview, product analysis, customer insights, and operational performance.

It also added interactive features like slicers and filters to enable dynamic exploration of data depending on the parameters like country, year, category. This design allowed the system to be able to process and visualize data efficiently and deliver meaningful insights by use of interactive dashboards.

4.2 Output Results

The project outcomes are reported in a form of a set of interactive dashboards, which offer information about different business aspects.

The overview dashboard gives highlights on the entire system performance by displaying that the total revenue made by the system is about 59.50 million, total orders made, average revenue per order, number of customers served and the number of units sold. The trend of monthly revenue shows that there are variations in revenue over the year where some months have significant peaks and others have decreases which hint to the seasonal variation in sales.

Analysis dashboard of products indicates that specific products are always bringing high revenue as indicated in the top 10 products chart. The units sold per category also show that there has been a fairly balanced performance of the company in terms of selling furniture, stationery, accessories, apparel, and electronics. The return distribution indicates that a large portion of the products is not returned, and only a minor per cent of them are returned.

The customer and sales insight dashboard shows that the revenue is spread over various countries with some countries providing higher revenue as compared to others. The analysis of the payment methods reveals that there is a relatively even balance of the various payment options used, including bank transfer, credit card and PayPal meaning that there is no high dependency on one payment method. The top customers chart is used to mark a few customers who earn a substantial amount of total revenue.

The operational dashboard offers information on the performance of logistics and delivery. The analysis of shipment providers indicates that the various providers have the same number of orders. The cost of average shipping among providers is stable, which means the cost of operation is steady. Moreover, order priority distribution has depicted almost equal distribution of high, medium and low order priorities.

The profitability analysis dashboard shows the correlation between revenues, costs, and profit based on various categories. The visualizations show that some of the categories have a higher contribution to the total profit, and there is evident difference in profitability across the categories. The waterfall chart also describes the contribution of each category to the overall profit whereas the breakdown analysis gives more information about the distribution of profit among categories, countries, and sales channels. This aids in determining the most lucrative areas of the business, and the cost-efficientness.

The customer segmentation dashboard offers information on customer behavior by classifying customers into high, medium and low value groups. The distribution reveals that there is a large percentage of customers in low and medium value category but the high value customers are fewer but have a greater percentage in total revenue. Analysis of behavior shows that customers who make more frequent orders have a high total spending. The segmentation assists in the interpretation of customer contribution and assists business specific strategies.

Time-series and growth analysis dashboard shows the movement of revenue over time in the moving average and percentage of growth measures. The moving average will smooth out all the short-term changes and show the general trend of revenue and the percentage change in revenue per month will indicate the fluctuations in performance and will indicate negative and positive growth periods. The category-wise trend analysis also shows the performance of various product categories over time, which can be used to determine seasonal trends and growth stability.

The advanced analytics and forecasting dashboard will offer proactive insights into the performance of the business. The revenue forecast visualization is a projection of the future trends of revenue according to the past information, which shows that there might be an increase or variation in revenue. The key influencers visual is used to determine the key factors that affect the level of revenue and the projected vs current revenue comparison gives the difference between the expected

and actual performance. Also, the scenario analysis feature will enable the adjustment of the price to see how it affects the estimated revenue, which will facilitate decision making.

4.3 Performance Analysis

The system can be rated on its effectiveness in processing data as well as producing meaningful insights in the form of visualization. Aggregation of data and real-time interaction using the Power BI is fast and the user can explore large data sets without major delays.

Analytically, the dashboards are quite good in bringing out key business metrics and trends. The system is able to determine the revenue patterns, the performance of the product, the customer contribution and the efficiency of the operations. The filters and slicers can be used to increase the usability of the system as the user can tailor the view and narrow down on particular portion of data.

The uniformity of the various categories and shipment providers is a clear sign of the stability of the business model, with even balances in the different aspects of performance. The minimal product returns are an indicator that there is good product quality or customer satisfaction. Moreover, the flexible nature is manifested in the balanced spread of the payment methods.

On the whole, the system is effective in data processing, visualization, and interaction between the user, which makes it a good business analysis and decision-making tool.

4.4 Discussion

The obtained analysis results give very important insights into the business operations and trends in performance. The project shows how data visualization tools such as Power BI can be used to transform raw data into useful insights to make decisions.

Among the most important things that can be observed is the necessity to find the best performing products and customers because they are the ones that drive the total revenue. On the same note, the knowledge on seasonal trends in revenue would enable business to plan inventory and marketing strategies better.

The comparison of the rates of returns and shipping costs also gives an understanding of the efficiency of the operations that could be utilized to streamline the logistic process and minimize unrelated costs. The even distribution of categories and method of payment means that the business structure is diversified and stable.

The implementation of machine learning models to forecast sales trends, identify anomalies, or predict the behaviour of customers can be improved in the future.

To sum up, the project is apparently a useful resource to comprehend and enhance the performance of the business, as it adequately illustrates how the data analytics and data visualization tools can be used to analyse business data and derive actionable insights.

Chapter 5: Project Conclusion and Presentation

5.1 Conclusion

The present project is a successful attempt to illustrate how data analytics and visualization can be used to analyse a dataset related to online sale with the help of Microsoft Power BI. Through the combination of preprocessing data in Google Colab and interactive dashboard creation in power BI, the project will offer an in-depth analysis of the business performance on various levels.

The main metrics that the eight dashboards, which include overview, product analysis, customer insights, and operational analysis, show, are revenue trends, product performance, customer contribution, and logistics efficiency. The data can be clearly understood through visual means such as charts, maps and KPI indicators.

In general, the project shows the significance of using data to make decisions since raw data are transformed into valuable information. It is a powerful tool to track and manage the business performance and be able to detect the patterns which may be used to plan and optimize the business.

5.2 Limitations

The project has some limitations in spite of its effectiveness. Though the project involves simple forecasting and scenario analysis, there are no sophisticated models of machine learning and very sophisticated predictive modes. The data employed is fixed, which means that it does not allow real-time analysis or the constant update of insights.

Moreover, there are variables of the dataset that have quite homogeneous distributions, which makes the variation in some visualizations less profound. Pre-processed data is also used in the project and therefore, any error in the original data may affect the outcome. In addition to this, the dashboards are not role-based, or can be deployed on a large scale to be used in enterprises.

5.3 Future Scope

To illustrate, sales forecasting models may be created and used to forecast the future growth or decline of revenues, whereas anomaly-detecting methods may be created and used to determine an unusual pattern in transactions.

It is possible to also implement real-time data integration so that the dashboards can be constantly monitored and updated. There are other features like customer segmentation, recommendation systems and profitability analysis that may give more insights. Moreover, the introduction of the dashboard on cloud platforms or its integration with web applications can enhance the accessibility and usability of the same to the broader audience.

5.4 References

- Microsoft Power BI Documentation
- Google Colab Documentation
- Online tutorials and learning resources on data visualization and Power BI
- Online sales dataset used for analysis